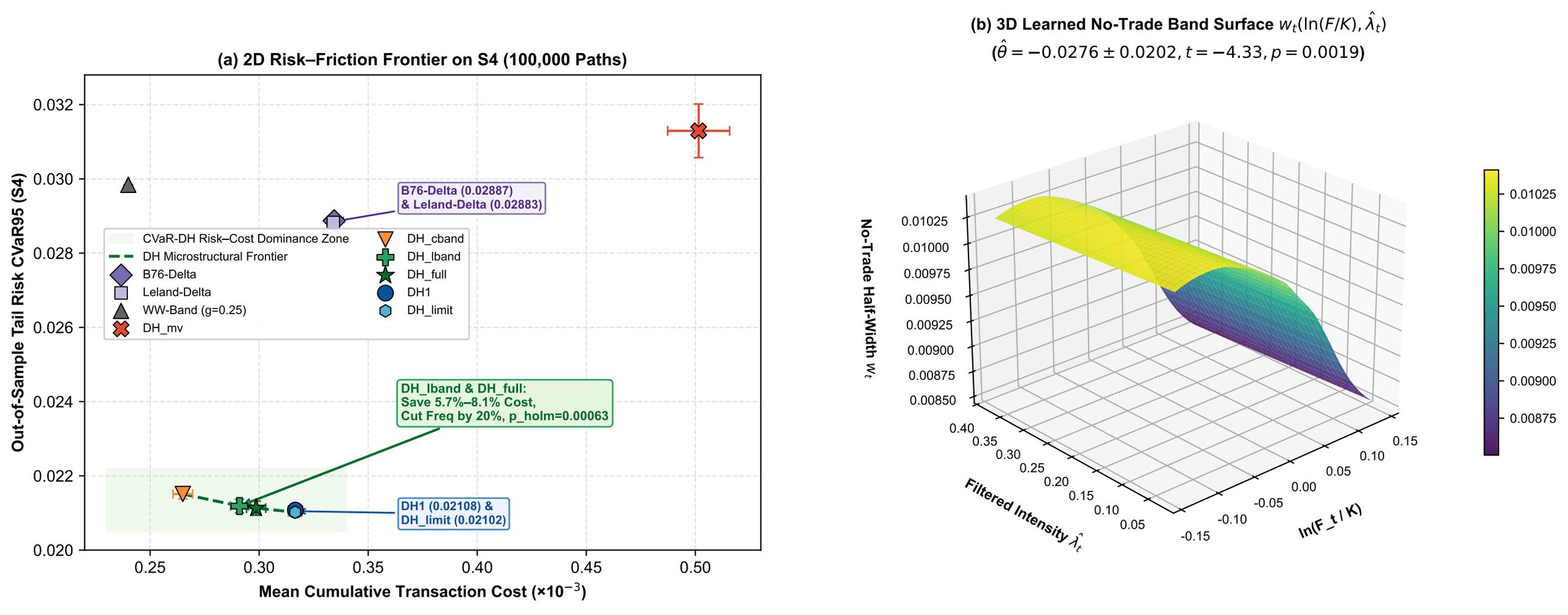

(a) 2D Risk–Friction Frontier on S4 (100,000 Paths)
Out-of-Sample Tail Risk CVaR95 (S4)
Mean Cumulative Transaction Cost (×10⁻³)
0.032
0.030
0.028
0.026
0.024
0.022
0.020
0.25
0.30
0.35
0.40
0.45
0.50
CVaR-DH Risk–Cost Dominance Zone
DH Microstructural Frontier
B76-Delta
Leland-Delta
WW-Band (g=0.25)
DH_mv
DH_cband
DH_lband
DH_full
DH1
DH_limit
B76-Delta (0.02887)
& Leland-Delta (0.02883)
DH_lband & DH_full:
Save 5.7%–8.1% Cost,
Cut Freq by 20%, p_holm=0.00063
DH1 (0.02108) &
DH_limit (0.02102)
(b) 3D Learned No-Trade Band Surface $w_t(\ln(F/K), \hat{\lambda}_t)$
($\hat{\theta} = -0.0276 \pm 0.0202, t = -4.33, p = 0.0019$)
No-Trade Half-Width w_t
0.01025
0.01000
0.00975
0.00950
0.00925
0.00900
0.00875
0.00850
Filtered Intensity $\hat{\lambda}_t$
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
ln(F_t / K)
0.15
0.10
0.05
0.00
-0.05
-0.10
-0.15